

Report R24 — what the eight rules of R23 protect

The strong-symmetry algebra, the conserved quantities, and why none of it is a code

Tier 1 analysis

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Abstract

R23 established that the six bounded dissipative rules relax onto a wall tiling with one period-2 dark qubit per wide tile, that W_{29}/W_{71} split a strong-symmetry block by a fair coin, and that W_{73}/W_{109} instead reach a hard-core constrained space carrying a Floquet blockade-Hadamard — PXP with the spin flip replaced by H . This report computes, exactly from the 4^N superoperator, the two algebras those statements live in: the strong-symmetry commutant $\mathcal{C}$ and the conserved-quantity space $\mathcal{F}$. Three results. *One*, the asymptotic algebra of every one of the eight is a *single* full matrix algebra M_D with trivial mixing factor, $\dim \mathcal{A} = D^2$ exactly, where D is the total number of recurrent basis states; exhaustively at $N = 4, 5$, precisely one Kraus operator is non-zero on the recurrent space and it is an isometry there, so the dissipation is transient-only and the asymptotic evolution is a closed unitary — not merely one decoherence-free subspace per attractor, but a single coherent block across all of them, with the coherences between distinct attractors surviving too. *Two*, R22's identity $\dim(\mathcal{F} \cap \mathcal{D}) = n_{\text{rec}}$ is a statement about the *monitored* chain; coherently only $n_{\text{wcc}} \leq \dim(\mathcal{F} \cap \mathcal{D}) \leq n_{\text{rec}}$ survives, and W_{29} at $N = 4$ realises the strict case — its fair coin is not a conserved quantity. Auditing R22's eight gap rules the same way splits them: five owe their gap entirely to the dephasing, while W_{36}, W_{44}, W_{100} keep a genuine coherent gap at $N = 5$. *Three*, none of the eight protects anything. Their code spaces are spanned by computational basis states, and we prove that such a space admits no subspace at all — of any dimension above one — that can so much as *detect* a single-site Z error. Distance one, no subcode helps, verified against the Knill–Laflamme conditions. What the eight are is a passive memory against one specific structured noise model, namely the machine's own resets, which annihilate the code space identically. The physics worth keeping is W_{73}/W_{109} : dissipation autonomously prepares a Rydberg-blockade Hilbert space of dimension $F(N+2)$ and then runs an entangling constrained Floquet circuit inside it, with preparation efficiency decaying like 0.9^N . That last statement has a one-line origin, which we prove: W_{73} and W_{109} are the unitary Floquet- $P^0 H P^0$ and Floquet- $P^1 H P^1$ rules W_{201} and W_{108} with a *single identity gate replaced by a reset*, and on the constrained space the reset can never fire, so the dissipative rule acts there exactly as its unitary parent — identically, to zero, for $N = 4, \dots, 11$.

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1 The two algebras, and why they differ

Throughout, $\{K_j\}$ are the Kraus operators of one brick-wall cycle, $\mathcal{D}$ is the algebra of operators diagonal in the computational basis, and

$$\mathcal{C} = \{A : [A, K_j] = 0 \ \forall j\}, \quad \mathcal{F} = \{A : \Phi^\dagger(A) = A\}.$$

$\mathcal{C}$ is the strong-symmetry algebra of Buča and Prosen [1]; $\mathcal{F}$ is the space of conserved quantities, the operational object, since $\text{Tr}(A\rho_t)$ is constant for every initial ρ exactly when $A \in \mathcal{F}$. Always $\mathcal{C} \subseteq \mathcal{F}$ — verified here to 3×10^{-14} — and $\mathcal{C}$ is an algebra while $\mathcal{F}$ in general is not: $\Phi^\dagger$ respects products only when it is a $*$ -automorphism, i.e. only in the closed case. Frigerio’s theorem [2] is the criterion: with a *faithful* stationary state $\mathcal{F}$ is a von Neumann algebra and equals $\mathcal{C}$. A large transient region is precisely what denies that hypothesis, which is why the eight rules below have $\dim \mathcal{C}$ far below $\dim \mathcal{F}$.

2 The measurement

Table 1: The eight rules of R23, obc0, exact from the 4^N superoperator. D is the dimension of the whole recurrent space, $\mathcal{A}$ the asymptotic algebra, $\dim \text{Fix}$ the fixed points of $\Phi^\dagger$ inside it. Every row has $\dim \mathcal{A} = D^2$.

rule	word	N	$\dim \mathcal{C}$	$\dim \mathcal{F}$	n_{wcc}	n_{rec}	D	$\dim \mathcal{A}$	$\dim \text{Fix}$
W_{28}	IIVD	4	25	45	6	6	9	81	45
W_{70}	IVID	4	32	45	6	6	9	81	45
W_{29}	EIVD	4	3	13	2	3	5	25	13
W_{71}	EVID	4	6	13	3	3	5	25	13
W_{157}	EIVI	4	3	13	3	3	5	25	13
W_{199}	EVII	4	5	13	3	3	5	25	13
W_{73}	VIID	4	19	35	4	4	13	169	35
W_{109}	EIIV	4	9	31	5	5	9	81	31
W_{28}	IIVD	5	58	106	8	8	14	196	106
W_{70}	IVID	5	58	106	8	8	14	196	106
W_{29}	EIVD	5	11	41	4	4	9	81	41
W_{71}	EVID	5	11	41	4	4	9	81	41
W_{157}	EIVI	5	18	41	4	4	9	81	41
W_{199}	EVII	5	18	41	4	4	9	81	41
W_{73}	VIID	5	33	96	6	6	24	576	96
W_{109}	EIIV	5	27	95	8	8	17	289	95

2.1 The dissipation is transient-only

Proposition 1 (Exhaustive at $N = 4, 5$). *For each of the eight rules, of the up to twelve Kraus operators of one cycle exactly one is non-zero on the recurrent subspace, and its restriction A there satisfies $\|A^\dagger A - \mathbb{K}\|_{\max} \leq 3.6 \times 10^{-15}$. Every other Kraus operator annihilates that subspace identically ($\|W^\dagger K W\| \leq 7.6 \times 10^{-15}$).*

Consequently the channel restricted to the recurrent space is a unitary, and the asymptotic algebra is the full M_D — which is what Table 1 reports as $\dim \mathcal{A} = D^2$, in every row.

This is stronger than “every attractor is a decoherence-free subspace”. That statement is per-enclosure; this one is joint. The recurrent space of W_{28} at $N = 4$ carries six terminal components of sizes 2, 2, 2, 1, 1, 1, and the channel decoheres neither within them *nor between them*: a superposition of states drawn from two different attractors is preserved exactly. It is also what makes $\mathcal{F}$ readable, because with a unitary U on the code space

$$\dim \mathcal{F} = \sum_i \text{mult}(\lambda_i)^2,$$

the sum over the multiplicities of U ’s eigenvalues. Table 1 can then be read spectroscopically, and R23 rev.2 has since diagonalised U directly, which both confirms the reading and corrects one guess made from the totals alone.

For the six bounded rules $\dim \mathcal{F} \approx D^2/2 - 45/81, 13/25, 106/196, 41/81$ — the signature of a two-fold spectrum. Diagonalising confirms it exactly: two distinct eigenvalues every time and $U^2 = \mathbb{K}$ to 2.2×10^{-16} , with multiplicities (6, 3) for W_{28} at $N = 4$ and (3, 2) for $W_{29}/W_{71}/W_{157}/W_{199}$ there, reproducing 45 and 13. So R23’s period-2 flicker, originally measured on trajectories, is forced by the spectrum.

For W_{73} at $N = 5$ the total $\dim \mathcal{F} = 96$ is right but its factorisation is *not* 6×4^2 , as a first reading of the totals suggested. U has twelve distinct eigenvalues with multiplicities (8, 4, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1), giving $64 + 16 + 4 + 4 + 8 = 96$; eight sit at exactly $+1$, four at -1 , and the rest are complex and off the roots of unity. W_{109} at $N = 5$ has $D = 17$ and multiplicities (9, 2, 2, 2, 1, 1), giving 95. The conclusion is unchanged and in fact sharpened — complex eigenvalues off the unit roots mean no period at all, $\|U^2 - \mathbb{K}\| = 0.5$ — but a multiplicity pattern must be read off the spectrum, never inferred from $\dim \mathcal{F}$.

2.2 Scope

Proposition 1 covers all enclosures at $N = 4$ and 5, which is exhaustive at those sizes but only those sizes. The independent check on the Tier-2 side (top three attractors per rule at $N = 11$, plus a basis-by-basis pass on the largest attractor of W_{73}/W_{109} at $N = 9, 11$) reaches further up but not across. Neither should be quoted as a theorem for all N .

3 The squeeze: what the monitoring was doing

R22 Proposition 4 states $\dim(\mathcal{F} \cap \mathcal{D}) = n_{\text{rec}}$. That is a statement about the *monitored* channel. For diagonal A the diagonal of $\Phi^\dagger(A)$ is exactly the population chain’s answer, so harmonicity is necessary; but $\Phi^\dagger(A)$ need not *be* diagonal, and membership of $\mathcal{F} \cap \mathcal{D}$ demands the off-diagonal part vanish as well. Hence

$$n_{\text{wcc}} = \dim(\mathcal{C} \cap \mathcal{D}) \leq \dim(\mathcal{F} \cap \mathcal{D}) \leq n_{\text{rec}}. \quad (1)$$

The left equality is exact for the coherent channel and holds in every row we have measured. The right inequality is saturated by construction for the monitored chain and can be strict otherwise.

W_{29} at $N = 4$ is the counterexample. Three terminal components, and $\dim(\mathcal{F} \cap \mathcal{D}) = 2$ — sharply, the smallest relative singular value of the off-diagonal block is 0.357, not a tolerance question. R23 identifies the mechanism: the third component is reached from a shared transient by a fair coin, and a fair coin is not a conserved quantity. The coherent channel simply has no observable that records which way it fell.

Auditing R22’s eight gap rules. The same computation applied to the rules that motivated R22 splits them in two (Table 2). Five have $\dim(\mathcal{F} \cap \mathcal{D}) = n_{\text{wcc}}$ at both sizes: their WCC/SCC gap is an artifact of dephasing and disappears when the monitoring is switched off. Three — W_{36}, W_{44}, W_{100} — keep a genuine coherent gap at $N = 5$.

Table 2: R22’s gap rules re-measured coherently, obc0. “monitored-only” means $\dim(\mathcal{F} \cap \mathcal{D})$ collapses to n_{wcc} although the graph reports $n_{\text{rec}} > n_{\text{wcc}}$.

rule	word	N	n_{wcc}	$\dim(\mathcal{F} \cap \mathcal{D})$	n_{rec}	verdict
W_{44}	IIDV	4	5	5	6	monitored-only
W_{104}	DIIV	4	3	3	4	monitored-only
W_{203}	VEII	4	1	1	3	monitored-only
W_{217}	VIEI	4	1	1	3	monitored-only
W_{233}	VIIE	4	3	3	4	monitored-only
W_{104}	DIIV	5	3	3	5	monitored-only
W_{203}	VEII	5	1	1	4	monitored-only
W_{217}	VIEI	5	1	1	4	monitored-only
W_{219}	VEEI	5	1	1	2	monitored-only
W_{233}	VIIE	5	3	3	4	monitored-only
W_{36}	IDDV	5	7	9	9	genuine gap
W_{44}	IIDV	5	7	8	9	genuine gap
W_{100}	IDIV	5	7	8	9	genuine gap

Two things this does *not* disturb. R22’s Theorem 1 concerns projections, and $\mathcal{C}$ and $\mathcal{F}$ contain the same projections, so the Boolean algebra of conserved questions still has exactly n_{wcc} atoms whichever channel is meant. And the measurements of R22 §3.4–3.5 are measurements of the monitored chain, which is the right object for the question they answer — what a repeatedly observed system remembers. The correction is one of scope, not of arithmetic.

4 Trivial, and not

4.1 The six are physical qubits with extra steps

$W_{28}, W_{70}, W_{29}, W_{71}, W_{157}, W_{199}$: R23 shows every enclosure is a subcube 2^k over a frozen wall backbone, so the decoherence-free space factorises as a tensor product of *single-site* dark qubits, each running H^t . There is no encoding: a logical qubit is one physical site. There is no non-locality, hence nothing for a distance to be built out of, and the operator entanglement of the asymptotic unitary is bounded, which is R23’s mechanism for the N -independent χ_{sat} .

$\dim \mathcal{C}$ for these is small and erratic — 3, 5, 6, 25, 32 at $N = 4$, and note that W_{157} and W_{199} have identical $\mathcal{F}$ but $\dim \mathcal{C} = 3$ against 5. That asymmetry is a fact about the *transients*, since $\mathcal{C}$ must commute with the Kraus operators everywhere, not only on the recurrent space. It is a reminder that $\mathcal{C}$ is not the right invariant for asymptotic questions.

4.2 W_{73} and W_{109} are the interesting ones

Their largest enclosure is the hard-core constrained space, of dimension $F(N+2)$ and $F(N)$, and the induced one-period map is R23's blockade-controlled Hadamard — PXP with the spin flip replaced by H , in Floquet rather than Hamiltonian form. Combining that with Proposition 1:

The dissipation autonomously prepares a Rydberg-blockade Hilbert space and then runs an entangling constrained Floquet circuit inside it.

That is a statement about dissipative state preparation of a kinetically constrained manifold, and it is the physics in this family worth reporting. Its efficiency is finite and decaying. The probability that a uniformly random initial state is absorbed into the largest enclosure, from the exact absorption probabilities:

N	4	5	6	7	8	9	10	11	12
W_{73}	0.625	0.563	0.500	0.438	0.398	0.344	0.316	0.272	0.251
W_{109}	0.313	0.438	0.281	0.328	0.234	0.254	0.190	0.199	0.151
dim	8	13	21	34	55	89	144	233	377

Roughly 0.9^N , against a target growing like φ^N . So the preparation is probabilistic and needs heralding; it is not a deterministic pump. Whether a modified rule can push this to 1 is the obvious question and is not attempted here.

5 Nothing here is a code

Theorem 1 (A basis-spanned code space has distance one). *Let the code space be spanned by computational basis states. Then no subspace of it of dimension ≥ 2 satisfies the Knill–Laflamme condition [3] for the single-site Pauli errors — not the correction condition, and not even the weaker detection condition.*

Proof. Let W be the isometry onto the code space and V that of a candidate subcode. Detection of Z_i requires $V^\dagger W^\dagger Z_i W V = c_i \mathbb{I}$, so the subcode lies inside an eigenspace of $M_i := W^\dagger Z_i W$. Because W is basis-spanned, each M_i is diagonal in the codeword basis with entries ± 1 , namely the i -th bit of each codeword. Distinct codewords are distinct bit strings, so the joint spectrum $\{(\pm 1)_i\}$ separates them and every joint eigenspace of $M_1, \dots, M_N$ is one-dimensional. Hence $\dim V \leq 1$. $\square$

R23 establishes that the asymptotic support of all eight rules is basis-spanned; we re-verify it at $N = 4, 5$ against the superoperator. So all eight have distance one and *no subcode helps*. Checked directly: the Knill–Laflamme condition fails at weight one for every one of them.

What is true is the complementary statement, and it is the same lesson R22 §8 drew from majority voting. The eight are perfectly protected against the machine's own dissipation — Proposition 1 says the jump operators annihilate the code space identically, an *infinite*-distance noiseless subsystem in the sense of Knill, Laflamme and Viola [4] for that one structured noise model — and completely unprotected against anything else. Fragmentation protects everything it finds, including the errors, which is why it protects nothing selectively.

One nuance worth recording, because it is the natural place to look for a positive result. The blockade constraint of W_{73} *does* make some bit-flip errors leave the code space, and those are detectable by a constraint check. But not all of them: $\dots 000 \dots \rightarrow \dots 010 \dots$ stays inside. The classical minimum distance of the hard-core space is 1, so even restricted to X errors the constraint is a detector of some errors, not a code.

5.1 Three exponents, and no gap anywhere

Beyond the sizes the superoperator reaches, the graph engine gives the growth of the protected space directly, since Prop. 1 makes D the number of recurrent basis states. For W_{73} and W_{109} , obc0, N up to 18:

quantity	growth	identification
n_{rec} , the number of enclosures	$\psi = 1.4656$	supergolden, A000930
$d_{\text{max}}^{\text{rec}}$, the largest enclosure	$\varphi = 1.6180$	Fibonacci, the blockade space
D , the whole protected space	1.8393	tribonacci constant

D for W_{73} runs 13, 24, 44, 81, 149, 274, 504, 927, 1705, 3136, 5768, ... — each term the sum of the previous three, ratio settled at 1.8393 by $N = 9$; W_{109} gives 9, 17, 31, 57, 105, 193, 355, ..., the same recurrence and the same constant. So the total decoherence-free dimension grows *faster* than any single enclosure: $\log_2 1.8393 = 0.879$ qubits per site protected in total, against $\log_2 \varphi = 0.694$ inside the largest blockade sector. The difference is exactly the cross-enclosure coherence that Prop. 1 says survives, and that a per-attractor account would miss.

The other column worth having: $n_{\text{wcc}} = n_{\text{rec}}$ at *every* N from 4 to 18, for both rules. These two are fragmented in both senses with no gap at any size we can reach — which is what a paper built on “fragmented in both senses, and non-unitary” needs, and which §3 shows is not automatic.

6 The dissipative pair is the unitary pair, with one letter changed

R23 identifies the induced unitary on the largest attractor of W_{73} and W_{109} numerically, as a brickwork of blockade-controlled Hadamards. There is a reason, and it is one line of gate arithmetic.

Read the gate words in slot order (00), (01), (10), (11), the slot being the state of the two neighbours:

$$W_{201} = \text{VIII}, \quad W_{73} = \text{VIID}; \quad W_{108} = \text{IIIV}, \quad W_{109} = \text{EIIV}.$$

W_{201} fires a Hadamard exactly when both neighbours are empty — it *is* the Floquet $P^0 H P^0$ model — and W_{108} fires when both are occupied, the $P^1 H P^1$ model. And each dissipative rule is its unitary parent with a *single* identity replaced by a reset: $W_{73} = W_{201}$ with slot (11) changed $I \rightarrow D$, and $W_{109} = W_{108}$ with slot (00) changed $I \rightarrow E$.

Proposition 2 (The reset is invisible on the constrained space). *Let $\mathcal{C}_N^1$ be the span of the basis states with no two adjacent 1s, and $\mathcal{C}_N^0$ likewise for 0s. Then $\mathcal{C}_N^1$ is invariant under the channel of W_{73} , no jump operator ever fires on it, and W_{73} acts there exactly as W_{201} . Dually, $\mathcal{C}_N^0$ is invariant under W_{109} , which acts there exactly as W_{108} .*

Proof. W_{73} ’s reset D sits in slot (11), so it can only fire at a site both of whose neighbours are 1; on $\mathcal{C}_N^1$ such a site must itself be 0, and D resets to 0, so it acts as the identity and its jump Kraus operator vanishes. The only non-trivial gate left is the Hadamard in slot (00), which can set a site to 1 only when both neighbours are 0, creating no adjacent pair; sites updated in the same brick-wall layer are two apart and their common neighbour is 0, so simultaneity does not spoil this. Hence $\mathcal{C}_N^1$ is invariant and the map is the no-jump Kraus operator alone, which is W_{201} ’s unitary. The argument for W_{109} is the same with 0 and 1, and E in place of D. $\square$

Measured, the two one-cycle matrices agree *identically* — not to machine precision, to exactly zero — for $N = 4, \dots, 11$, with no amplitude leaving the subspace, only the empty Kraus label occurring, and $\|M^\top M - \mathbb{K}\| \leq 2.3 \times 10^{-16}$.

So the mechanism behind “some dissipative models inherit fragmentation” is not an analogy. The reset does no work at all in the steady state; its entire role is to *drive the system into* the constrained subspace, where the dynamics is the unitary model itself. That also explains R23’s $a = 0$ versus $a = 1$ blockade: the projector is inherited from whichever slot the parent’s Hadamard occupies.

Two caveats. The proposition gives invariance, not attraction — that $\mathcal{C}_N^1$ is reached is a separate fact, and §3’s table shows it is reached only with probability $\approx 0.9^N$. And $\mathcal{C}_N^1$ has dimension $F(N + 2)$ while R23 reports $F(N)$ for W_{109} ’s largest *attractor*: the obc0 boundary counts sites past the chain end as 0, which for W_{109} lets the reset fire at the ends, so the invariant space splits further there. Invariance of the full $\mathcal{C}_N^0$ is what is proved; irreducibility of it is not.

7 What this leaves for the PRL

Three things the above supplies for a paper built on “fragmented, in both senses, and non-unitary”.

1. **The mechanism sentence, and it is better than expected.** The dissipative rules inherit fragmentation because their dissipation is *transient-only*: asymptotically the channel is a unitary on an exponentially large constrained subspace (Prop. 1). For the two rules that matter it is sharper still — W_{73} and W_{109} are the Floquet- P^0HP^0 and Floquet- P^1HP^1 models with a *single identity replaced by a reset*, and on the constrained space the reset provably never fires (Prop. 2). The dissipative models do not merely resemble the unitary ones; they *are* them, plus a pump that drives the system into the constrained sector. If the paper introduces Floquet-PHP and Floquet-PHQ as its two unitary models, then Fig. 3’s dissipative counterparts are the same two words with one letter changed, which is about as compact as a third figure can get. That is also why an SCC census suffices there, and why deferring the WCC/SCC distinction to a companion paper is defensible: for seven of the eight, $n_{\text{wcc}} = n_{\text{rec}}$ at every size we can check coherently.
2. **One caveat that must be handled before the figure is drawn.** It is W_{29} , and now $W_{36}/W_{44}/W_{100}$ from §3: the SCC count is not always a coherent invariant. A census presented as “fragmentation” should either restrict to rules where (1) is tight or say which quantity is plotted.
3. **A gap in the evidence, and a trap in closing it.** R23 reports χ_{sat} saturating *in N* for the six and growing exponentially for W_{73}/W_{109} . A headline about linear growth *in time* is a different axis, and the dissipative rules have not been measured on it. The trap is that the obvious protocol is self-defeating: Proposition 1 says there is no dissipation *inside* an enclosure, so a $\chi(t)$ run started from a state already in the largest enclosure measures the operator entanglement of a purely unitary constrained circuit and cannot support any claim about the dissipative model. The dissipative content lives entirely in the transient. Whatever protocol is chosen must therefore start outside the enclosure and pay for the transient. A second, practical obstacle: the superoperator route OOMs for W_{73} at $N = 7$, $t = 5$ (27 GB, because the zip-up materialises an untruncated bond at $16\chi \approx 12000$), so those two need a contraction that never forms that bond. $\chi(t)$ for the six bounded rules is runnable today.

Reproduce

Everything is in `quantum/asymptotic.py`, which builds the dense 4^N superoperator through `quantum/peripheral.py`. The commutant is intersected one Kraus operator at a time rather than as a single stack — with a reset at every site the stack has $2^N d^2$ rows and exhausts memory at $N = 5$.

```
python -m qca_fragmentation.quantum.asymptotic --rules 28,29,73,109 --N 4
pytest qca_fragmentation/tests/test_asymptotic.py # 147 tests
```

The tests pin Table 1 row by row, check Proposition 1 for all eight at $N = 4, 5$, confirm the code space is basis-spanned, verify Theorem 1 through the Knill–Laflamme condition, and pin both sides of (1) including the W_{29} counterexample and the W_{36} -versus- W_{203} split.

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